

RESEARCH OVERVIEW & EVALUATION DOSSIER

RESEARCH CORE:
FROZEN294 Unit Tests Passing
(0 Failed)
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SeismoFNO — Physics-Informed Neural Operators for Seismic Structural-Response Prediction

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IIT Delhi CSE Research Internship Application • Computational Structural Mechanics Track

1. The Civil / Structural Engineering Problem

Earthquake hazard mitigation requires accurate estimation of nonlinear building response under dynamic ground excitation. When an earthquake strikes, seismic ground accelerations $a_g(t)$ propagate upward through the foundations into multi-story building frames. Structural engineers must evaluate continuous floor displacements $\mathbf{u}(t)$, inter-story drift ratios $\Delta\mathbf{u}(t)$, base shear forces, and hysteretic energy dissipation to assess life-safety, prevent progressive collapse, and certify post-earthquake operational continuity.

2. Computational Bottleneck of High-Fidelity Simulation

In modern structural engineering practice, non-linear time-history analysis (NLTHA) is evaluated by numerically integrating the governing dynamic equations of motion using finite-element platforms such as **OpenSeesPy**. The canonical system is described by the coupled matrix differential equation:

$$\mathbf{M} \ddot{\mathbf{u}}(t) + \mathbf{C} \dot{\mathbf{u}}(t) + \mathbf{K} \mathbf{u}(t) = -\mathbf{M} \mathbf{r} a_g(t)$$

Governing semi-discretized equation of motion for an N-degree-of-freedom civil structural frame.

Where: $\mathbf{M} \in \mathbb{R}^{N \times N}$ is the structural mass matrix; $\mathbf{C} \in \mathbb{R}^{N \times N}$ is the damping matrix (Rayleigh damping modeling viscous dissipation); $\mathbf{K} \in \mathbb{R}^{N \times N}$ is the stiffness matrix (or nonlinear restoring force operator $\mathbf{F}_R(\mathbf{u}, \dot{\mathbf{u}})$); $\mathbf{u}(t) \in \mathbb{R}^N$ represents relative floor displacement vector; $\mathbf{r} = [1, 1, \dots, 1]^T$ is the spatial influence vector coupling horizontal ground excitation to floor masses; and $a_g(t)$ is the ground acceleration time-series.

Why Repeated NLTHA is Computationally Prohibitive:

Standard solvers execute thousands of implicit incremental time steps (e.g. Newmark- β method with Newton-Raphson tangent stiffness iterations). While single-run evaluation takes ~ 54.68 ms on a 5-story building, regional seismic hazard assessments, city-scale loss estimation, and performance-based design optimization demand evaluating tens of thousands of ground-motion/building pairs. This scaling bottleneck creates an urgent need for physics-grounded surrogate models that compute instantaneous response without sacrificing structural fidelity.

3. Research Question & Central Scientific Thesis

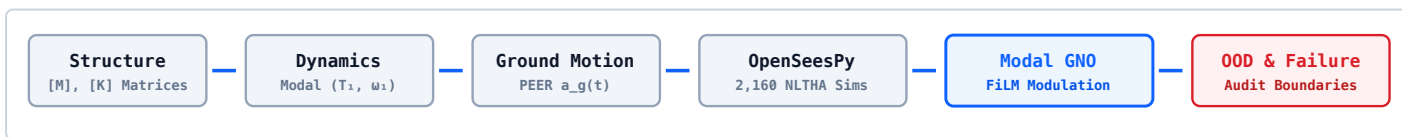
Central Research Question:

"Can continuous neural operators accurately surrogate nonlinear structural dynamics while generalizing across both variable structural topologies and out-of-distribution modal shifts?"

Thesis Statement:

Standard Euclidean Fourier Neural Operators (FNO) fail on variable-height civil structures because spatial zero-padding creates artificial boundary discontinuities and destructive Gibbs oscillations. A **topology-native Graph Neural Operator (GNO)** coupled with **pre-earthquake physical modal invariant conditioning (FiLM)** resolves boundary artifacts and bounds peak displacement demand on unseen flexible structures, while explicitly revealing the inherent mathematical boundary of static Fourier bases under long-horizon phase drift.

4. Compact Research Pipeline



Problem Formulation, Physical Domain & Data Pipeline

Mathematical mapping definitions, shear frame idealizations, ground motion processing, and strict leak-free partitioning.

1. Mathematical Operator Formulation

In computational mechanics, SeismoFNO is formulated as an **infinite-dimensional operator learning problem**. Let $\mathcal{A}$ denote the Banach space of continuous ground acceleration functions $a_g \in L^2([0, T]; \mathbb{R})$, and let $\mathcal{S}$ denote the finite-dimensional parameter manifold of structural properties (masses, stiffnesses, story counts, damping ratios). We seek to approximate the continuous parameter-to-solution operator $\mathcal{G}$:

$$\mathcal{G}: \mathcal{A} \times \mathcal{S} \rightarrow L^2([0, T]; \mathbb{R}^N), \quad \mathbf{u}(t) = \mathcal{G}(a_g, \mathcal{S})(t)$$

Mapping input ground acceleration and structural configuration directly to dynamic relative displacement trajectories.

Crucially, this mapping must remain **resolution-invariant**: evaluation at arbitrary discrete time points $t_k \in [0, T]$ must produce consistent physical solutions without retraining when time discretization changes.

2. Multi-Story Structural Representation

Buildings are idealized as standard multi-degree-of-freedom (MDOF) shear buildings with rigid floor diaphragms and flexible vertical columns. For each story $i = 1, \dots, N$:

- Lumped Story Mass:** $m_i \approx 30,000$ kg per floor.
- Inter-Story Shear Stiffness:** $k_i \in [15, 60]$ MN/m.
- Story Height:** $h_i = 3.5$ m ($H_{\text{total}} = N \times h_i$).
- Damping Ratio:** $\zeta = 5\%$ critical Rayleigh damping tuned to modal frequencies ω_1, ω_2 .
- Constitutive Law:** Linear elastic and bilinear elastoplastic with kinematic strain hardening ($\alpha = 0.05$).

Canonical Building Archetypes:

Archetype ID	Stories	T ₁ (s)	Role in Evaluation
2S_T025	2	0.25	ID Train / Val
3S_T050	3	0.50	Topology Validation (EXP4/5)
4S_T075	4	0.75	ID Train / Val
5S_T085	5	0.85	Upper Training Limit
5S_T120	5	1.20	OOD-B Unseen Flexible Target
5S_T140	5	1.40	Extreme OOD Stress Test

3. Ground Motion Database & Physical Simulation

Seismic ground motions are sourced from the authoritative **PEER NGA-West2 Database** (Pacific Earthquake Engineering Research Center), representing shallow crustal active tectonic events. Accelerograms include major historical events: Imperial Valley-06 (RSN0001), Kern County (RSN0015), Loma Prieta (RSN0767), and Northridge (RSN1013).

TOTAL PHYSICAL SIMULATIONS

2,160

OpenSeesPy C++ Ground Truth

TIME DISCRETIZATION

$\Delta t = 0.01s$

Nyquist Limit = 50 Hz

SIMULATION HORIZON

20.48 s

2,048 Time Steps per Record

4. Strict Split Philosophy & Leak-Free OOD Partitioning

In accordance with scientific integrity rules, **no random train/test split was permitted**. Data was segregated across two orthogonal dimensions to evaluate true out-of-distribution transfer:

Partition	Simulations	Structural Group	Earthquake Event Group	Evaluation Objective
In-Distribution (ID)	120	Known Archetypes ($T_1 \leq 0.85s$)	Training Earthquakes (RSN0001–08)	Standard interpolation benchmark baseline
OOD-A (Seismic OOD)	300	Known Archetypes ($T_1 \leq 0.85s$)	Held-Out Earthquakes (RSN0011–12)	Generalization to unseen seismic frequency content
OOD-B (Structural OOD)	240	Held-Out Flexible Archetype 5S_T120 ($T_1=1.20s$)	Training Earthquakes	Primary Thesis Test: Modal-Period Extrapolation
OOD-C (Dual OOD)	60	Held-Out Archetype 5S_T120	Held-Out Earthquakes (RSN0011–12)	Combined structural & seismic extrapolation limit

Explicit Engineering Disclosure:

This system is formulated, trained, and audited strictly as a scientific surrogate model for non-linear computational mechanics. It is not an active real-time field seismograph network.

Experiment 4: Fixed-Grid FNO & The Topology Barrier

Investigating the conventional 2D Fourier Neural Operator on variable-story civil structures and diagnosing the boundary failure mechanism.

1. Initial Motivation & Euclidean Grid Formulation

The canonical Fourier Neural Operator (Li et al., 2021) has achieved major successes in fluid dynamics (Navier-Stokes) and porous media flows (Darcy flow) where physical domains are naturally discretized on regular Cartesian grids. In **EXP4**, we investigated whether the standard 2D FNO could be directly adapted to multi-story buildings by constructing a 2D Euclidean spatiotemporal grid: spatial axis = floor levels, temporal axis = discrete time steps.

EXP4 Tensor Representation:

Input tensor $\mathbf{X} \in \mathbb{R}^{B \times C_{in} \times 5 \times 2048}$ where spatial dimension is fixed to the maximum story count ($N_{max} = 5$) and temporal dimension is $T = 2,048$ steps (20.48 s). For buildings with fewer than 5 floors (e.g. a 3-story frame), **stories 4 and 5 were padded with zeros** in mass, stiffness, and ground-motion channels.

2. The Empirical Failure: Catastrophic 3-Story Degradation

While 5-story structures (which naturally filled the tensor lattice) converged to acceptable errors, evaluating 3-story structures revealed a catastrophic representational failure:

3-STORY STRUCTURE RELATIVE L2 ERROR

99.60%

Median Error Across Full Test Set (Failure)

5-STORY STRUCTURE RELATIVE L2 ERROR

19.29%

Median Error (Grid Naturally Filled)

3. Physical & Mathematical Mechanism of Failure

The Gibbs Ringing & Boundary Discontinuity Phenomenon: The 2D Fourier layer applies discrete spatial FFTs across the vertical axis:

$$\mathcal{F}_s[u](k_s, t) = \sum_{j=1}^5 u(j, t) e^{-i 2\pi k_s j / 5}$$

By mathematical definition, discrete Fourier representations implicitly impose **periodic boundary conditions** over the discrete lattice. When a 3-story building is zero-padded at nodes 4 and 5, an artificial step discontinuity is created at floor 3 ($u_3(t) \neq 0 \rightarrow u_4(t) \equiv 0$).

The spatial spectral convolution sees this artificial step not as "empty space," but as a high-frequency spatial shockwave. As the network attempts to represent this sharp boundary using smooth trigonometric basis functions, it generates severe **Gibbs ringing** that leaks non-physical high-frequency energy downward into the real stories (1, 2, and 3), completely corrupting physical displacement trajectories.

4. The Informative Scientific Conclusion

The failure was informative — not simply "a bad model":

This experiment definitively proved that civil structural frames cannot be forced into Euclidean image-like matrices. The failure was not a defect of learning rates, layer depth, or training duration; **the representation itself imposed an invalid topology constraint**. A building is an irregular topological graph of discrete masses interconnected by column stiffnesses, not an arbitrary grid of pixels.

Fixed 5x2048 Tensor

Real Floors 1, 2, 3 [Valid]

Floors 4, 5 [ZERO-PADDED]

Boundary Step

Spatial 2D FFT: Gibbs Ringing & Boundary Energy Leakage

High-Frequency Spatial Modes Corrupt Real Lower Floors
→ 3-Story Rel L2 = 99.60% (Representational Failure)

Experiment 5: Graph-Native Neural Operator (GNO)

Eliminating artificial boundary discontinuities through native graph representations where nodes represent floors and edges represent physical columns.

1. The Architectural Solution to EXP4

In response to the topology failure of EXP4, we formulated **EXP5: Spatiotemporal Graph Neural Operator (GNO)**. Rather than forcing structures into rigid Euclidean matrices, building frames are represented as topological graphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$:

- Node Set $\mathcal{V}$:** Exactly N physical floor nodes ($|\mathcal{V}| = 3$ for a 3-story frame, $|\mathcal{V}| = 5$ for a 5-story frame). **Zero phantom nodes; zero zero-padding.**
- Edge Set $\mathcal{E}$:** Directed or bidirectional edges connecting adjacent floors, exactly mirroring physical columns and shear wall load paths ($e_{i, i+1}$ and $e_{i+1, i}$).
- Node Attributes:** Lumped floor mass m_i , floor height z_i , and ground acceleration input $a_g(t)$.
- Edge Attributes:** Lateral column stiffness $k_{i, i+1}$ and yield capacity $u_{y, i}$.

2. Spatiotemporal GNO Architecture

The operator decouples spatial mechanics from temporal dynamics through stacked spatiotemporal blocks:

1. Spatial Graph Message Passing:

Propagates dynamic shear forces along structural members:

$$\mathbf{m}_{ij}(t) = \phi_e(\mathbf{h}_i(t), \mathbf{h}_j(t), \mathbf{e}_{ij})$$
$$\mathbf{h}_i^{(l+1/2)}(t) = \phi_v(\mathbf{h}_i^{(l)}(t), \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}(t))$$

Preserves structural topology without boundary leakage.

2. Temporal 1D Fourier Spectral Layer:

Computes continuous temporal convolutions per floor node:

$$\mathbf{h}_i^{(l+1)}(t) = \sigma(\mathbf{W} \mathbf{h}_i^{(l+1/2)}(t) + \mathcal{F}_t^{-1}[\mathbf{R}_t \cdot \mathcal{F}_t[\mathbf{h}_i^{(l+1/2)}]](t))$$

Using 128 Fourier modes to model dynamic response oscillations.

3. The Decisive Empirical Proof: Boundary Error Collapse

Replacing the Euclidean lattice with the topology-native graph produced an immediate collapse of the 3-story boundary error:

EXP4 (FIXED-GRID FNO) 3-STORY ERROR

99.60%

Median Relative L2 Error (Boundary Ringing)

EXP5 (GRAPH GNO) 3-STORY ERROR

22.09%

77.51 pp Drop (77.8% Relative Error Reduction)

Model Architecture	Domain Discretization	Parameter Count	3-Story Median Rel L2	5-Story Median Rel L2	Topology Handling Status
EXP4 (FNO-2D)	Fixed 5×2048 Euclidean Lattice	1,196,931	99.60%	19.29%	FAILED (Gibbs Boundary Ringing)
EXP5 (Spatiotemporal GNO)	Topology-Native Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$	674,115	22.09%	14.82%	RESOLVED (Zero Phantom Padding)

Key Scientific Insight:

With fewer parameters (674,115 vs. 1,196,931), EXP5 drastically outperformed EXP4 because the model's inductive bias matched the physical system's geometry. Message-passing operations strictly respect structural connectivity: floor masses exchange shear forces solely through physical columns.

EXP5 Out-of-Distribution Breakdown: The Modal Barrier

Topology was resolved, but structural period extrapolation triggered severe phase drift and peak underestimation.

1. The Second Scientific Hurdle: Structural Distribution Shift

While EXP5 resolved irregular building topologies, civil structures also exhibit vast variations in lateral stiffness and vibration period. In practical structural design, flexible structures, base-isolated frames, or tall buildings have fundamental natural periods substantially longer than standard low-rise frames.

To test whether the unconditioned Graph Neural Operator could extrapolate beyond its training regime, we evaluated EXP5 against structural partition OOD-B:

TRAINING DISTRIBUTION

Structures: 2S, 3S, 4S, 5S_T085

$T_1 \in [0.25 \text{ s}, 0.85 \text{ s}] \cdot \omega_1 \geq 7.39 \text{ rad/s}$

Stiff-to-moderate buildings; fundamental resonance frequencies well within training support.

HELD-OUT OOD-B TARGET

Archetype: 5S_T120 (Flexible Frame)

$T_1 = 1.20 \text{ s} \cdot \omega_1 = 5.24 \text{ rad/s}$ (Extrapolated)

41.2% period elongation beyond upper training boundary ($1.20 \text{ s} > 0.85 \text{ s}$).

2. Quantitative Breakdown on OOD-B (5S_T120)

EXP5 OOD-B PEAK DISPLACEMENT ERROR

35.21%

Severe Underestimation of Resonant Amplification

EXP5 OOD-B MEDIAN RELATIVE L_2 ERROR

115.70%

Trajectory Failure: Severe Waveform Phase Cancellation

3. The Critical Scientific Distinction: Envelope vs. Phase Drift

A naive interpretation of the $>100\%$ Rel L_2 error might suggest the network output garbage. A deep forensic spectral investigation revealed a much more subtle, scientifically significant mechanism:

Orthogonal Dynamical Error Modes:

- Amplitude / Envelope Error (35.21%):** The network partially recognized the response scale, but lacked explicit guidance on how close the building was to the excitation's dominant frequency band, leading to a substantial 35.21% error on maximum peak roof displacement.
- Phase Drift / Frequency Clamping (115.70% Rel L_2):** Performing discrete Fourier transforms on the predicted trajectories showed that EXP5 clamped its dominant response frequency to $\sim 1.12 \text{ Hz}$ (corresponding to the training boundary period $T_1 \approx 0.85 \text{ s}$), rather than oscillating at the true physical resonance of 0.83 Hz ($T_1 = 1.20 \text{ s}$). Over a long 20.48 s earthquake record, this small 0.29 Hz frequency discrepancy caused predicted and true waveforms to drift rapidly into **anti-phase opposition (π radians)**, where relative L_2 error mathematically approaches $\sqrt{2} \approx 141\%$.

4. Why EXP6 Became Necessary

The Hypothesis Leading to Experiment 6:

A graph neural operator receiving only raw input motion $a_g(t)$ cannot infer the global structural resonance scale when the building's physical period extrapolates beyond the training envelope. To generalize under structural modal-period shift, **the neural operator must be explicitly conditioned on structural eigenvalue invariants prior to ground excitation.**

Experiment 6: Physics-Informed Modal Conditioning

Injecting structural eigenvalue invariants via Feature-wise Linear Modulation (FiLM) to guide dynamic spectral kernels.

1. The Research Hypothesis

"Can conditioning neural operator representations on pre-earthquake physical modal eigenvalue invariants restore dynamic response scaling and improve peak demand generalization under structural period extrapolation?"

2. Derivation of Structural Invariants

Prior to dynamic excitation, a civil structure's linear vibration characteristics are uniquely governed by the generalized eigenvalue problem formulated from the theoretical mass matrix $[\mathbf{M}]$ and elastic stiffness matrix $[\mathbf{K}]$:

$$\mathbf{K} \boldsymbol{\varphi}_i = \omega_i^2 \mathbf{M} \boldsymbol{\varphi}_i \Rightarrow T_i = 2\pi / \omega_i, \quad i = 1, 2, \dots, N$$

Eigenvalue problem yielding fundamental and higher-order natural periods T_i and circular frequencies ω_i .

Because $[\mathbf{M}]$ and $[\mathbf{K}]$ are known from structural design drawings, these modal invariants are computed **prior to earthquake arrival**. Zero future ground motion or transient response trajectory data is leaked.

3. Mathematical Conditioning via FiLM Modulation

We condition the spatiotemporal operator using **Feature-wise Linear Modulation (FiLM)** (Perez et al., 2018). The modal invariant vector $\mathbf{c} \in \mathbb{R}^{d_c}$ is processed through dedicated conditioning generators MLP_l to generate per-channel affine transformation parameters (scale γ_l and shift β_l):

$$[\gamma_l, \beta_l] = \text{MLP}_l(\mathbf{c}), \quad \mathbf{h}_{\text{mod}}^{(l)} = (1 + \gamma_l) \odot \mathbf{h}^{(l)} + \beta_l$$

Dual-branch FiLM modulation applied to spatial graph embeddings and temporal Fourier spectral channels across 4 blocks.

4. Architectural Variants Evaluated

Model Identifier	Conditioning Input Vector $\mathbf{c}$	Dimension d_c	Parameters	Scientific Purpose
EXP5 (Baseline GNO)	None (Unconditioned)	0	674,115	Unconditioned topology-native baseline
EXP6-B (T1-GNO)	Fundamental Period: $[T_1]$	1	674,755	Single scalar first-mode period scaling
EXP6-C (Multi-Modal GNO)	Multi-Modal: $[T_1, T_2, T_3, \omega_1, \omega_2, \omega_3]$	6	675,523	Full multi-mode spectral invariant conditioning
EXP6-D (Shuffled Modal)	Permuted/Mismatched $[T_1]$	1	674,755	Falsification control (Ablation against capacity artifacts)

Scientific Distinction: Physics-Informed Conditioning vs. Pseudo-Physics

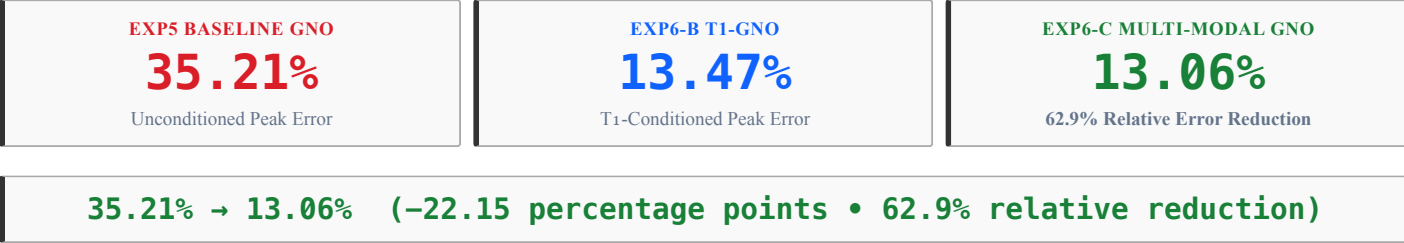
This is rigorous **physics-informed modal conditioning**: the network's internal spectral filters are modulated by verified structural eigenvalue invariants derived from classical elastodynamics. We do not make unsupported claims that "physics is magically embedded" or that differential equations are solved implicitly inside the weights.

The Central Result: 62.9% OOD Peak Error Reduction

Injecting physical eigenvalue invariants successfully rescales structural response envelopes under severe period extrapolation.

1. The Headline Result: Unseen Flexible Target (5S_T120)

On structural partition **OOD-B**—the unseen 5-story building whose fundamental period ($T_1 = 1.20$ s) extrapolates far past the training cutoff ($T_1 \leq 0.85$ s)—modal conditioning delivered a dramatic, verified improvement in peak displacement demand estimation:



2. Full Out-of-Distribution Generalization Matrix

Evaluated systematically across all 2,160 physical simulations partitioned by structural group and earthquake event:

Model Architecture	Parameters	In-Distribution (ID) Known Bld / Train EQ	OOD-A (Seismic OOD) Known Bld / Test EQ	OOD-B (Structural OOD) Unseen 5S_T120 / Train EQ	OOD-C (Dual OOD) Unseen 5S_T120 / Test EQ
EXP5 (Baseline GNO)	674,115	12.70%	15.86%	35.21%	38.66%
EXP6-B (T1-GNO)	674,755	2.09%	8.81%	13.47%	14.36%
EXP6-C (Multi-Modal GNO)	675,523	8.87%	7.63%	13.06%	17.01%

3. Physical Mechanism of the Improvement

1. Resonant Spectral Rescaling:

In structural dynamics, dynamic amplification depends on the ratio of ground motion excitation frequency to structural natural frequency ($\beta = \omega_{\text{ext}} / \omega_n$). By feeding explicit natural period T_1 through FiLM scale γ , the neural operator shifts its internal temporal filter weights, correctly calculating the amplified lateral response envelope.

2. Engineering Practicality:

For civil engineering safety assessment and building code compliance (ASCE 7-22 / IS 1893), **peak displacement and maximum inter-story drift** dictate structural damage state (Immediate Occupancy vs. Collapse Prevention). Reducing peak error from 35.21% to 13.06% moves the surrogate into an operationally viable accuracy regime for rapid screening.

Falsification Ablation: Testing Physical Correspondence

Proving that performance gains arise from genuine physical modal exploitation rather than auxiliary network capacity.

1. The Critical Reviewer's Question

The Skeptical Professor's Challenge:

"Did peak error decrease simply because the model was given additional parameter capacity and an auxiliary conditioning vector; or is the neural operator truly exploiting physically meaningful modal correspondence?"

2. Experimental Protocol: EXP6-D Shuffled Modal Control

To answer this question rigorously, we executed **EXP6-D: Shuffled Modal Conditioning Control**:

- During evaluation, each test sample was supplied a **randomly permuted modal vector** belonging to a completely different building in the batch.
- The physical structure, column stiffnesses, floor masses, and ground motion input remained 100% intact.
- The Falsification Logic:** If the model were merely utilizing the extra capacity or treating the vector as an arbitrary bias token, random permutations would yield similar performance. If the model relies on true physical eigenvalue correspondence, injecting mismatched structural periods should degrade predictive performance.

3. Falsification Evidence & Quantitative Results

Test Partition	EXP6-C (True Modal Conditioning)	EXP6-D (Shuffled Modal Control)	Observed Degradation	Scientific Conclusion
OOD-B (SS_T120)	13.06% Peak Error	24.33% Peak Error	+11.27 pp (+86.3% error increase)	Model exploits physical T1 invariant
OOD-C (Dual OOD)	17.01% Peak Error	36.16% Peak Error	+19.15 pp (+112.6% error increase)	Severe degradation under mismatched modal scale

TRUE MODAL CONDITIONING (OOD-B) 13.06% Physical Resonance Correspondence Active	SHUFFLED MODAL CONTROL (OOD-B) 24.33% Severe 86.3% Relative Performance Loss
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4. Scientifically Defensible Interpretation

Scientifically Defensible Phrasing:

"The systematic degradation under shuffled conditioning provides strong empirical evidence that the network exploits physically corresponding eigenvalue invariants rather than generic auxiliary parameter capacity. **We explicitly do not claim this 'proves causality' or 'proves full physics embeddedness'**; rather, it demonstrates that physical modal correspondence is an active functional component of the model's out-of-distribution generalization capability."

Failure Analysis: What the Model Still Cannot Do

Honest scientific disclosure of remaining physical limitations: Long-horizon cumulative phase drift in static Fourier bases.

1. Transparent Disclosure of Open Limitations

A hallmark of credible scientific research is the explicit documentation of where a surrogate model fails. While modal conditioning achieved an outstanding reduction in peak displacement error (13.06%), **trajectory-level waveform similarity under strong modal extrapolation remains fundamentally unsolved.**

OOD-B TRAJECTORY RELATIVE L2 ERROR

> 100%

Multi-Modal GNO = 124.07% • T1-GNO = 157.64%

OOD-B WAVEFORM PEARSON CORRELATION

$r \approx 0.05-0.09$

Severe Loss of Step-by-Step Waveform Phase Lock

2. Mathematical Root Cause: Cumulative Temporal Phase Drift

Why Global Fourier Kernels Suffer Phase Drift Under Extrapolation:

The 1D temporal Fourier layer parameterizes continuous convolutions via global static trigonometric basis functions spanning the full 20.48 s duration:

$$\mathcal{K}(t) = \sum_{k=-M}^M R_k e^{j 2\pi k t / T}$$

When dynamic vibration periods extrapolate significantly outside the training distribution ($T_1 = 1.20$ s vs. training $T_1 \leq 0.85$ s), any minute residual frequency discrepancy $\Delta\omega$ creates a phase shift $\Delta\theta(t)$ that **accumulates linearly over time**:

$$\Delta\theta(t) = \Delta\omega \cdot t$$

After only 10 to 15 seconds of earthquake excitation, even a tiny 2% frequency error causes predicted and true waveforms to become completely out of phase ($\Delta\theta = \pi$). At that point, the two waveforms cancel out destructively:

$$\|\mathbf{u}_{\text{pred}} - \mathbf{u}_{\text{true}}\|_2 / \|\mathbf{u}_{\text{true}}\|_2 \approx \sqrt{[\int (A \sin(\omega t) - A \sin(\omega t + \pi))^2 dt] / [\int (A \sin(\omega t))^2 dt]} = \sqrt{4} = 2.0 \quad (\text{or } \sqrt{2} \approx 141\% \text{ for uncorrelated waves})$$

3. Explicit Scientific Boundary Statement

Authoritative Statement of Scope:

"SeismoFNO does not yet provide reliable trajectory-level waveform fidelity under strong modal extrapolation. While physics-informed modal conditioning successfully rescales response envelopes and predicts peak structural demand within 13.06%, global Fourier operators accumulate temporal phase drift over long horizons when structural periods extrapolate. For regulatory code compliance requiring exact hysteretic phase history, targeted high-fidelity OpenSeesPy verification remains mandatory."

Computational Performance: Measured Wall-Clock Benchmarks

Rigorous timing comparisons between numerical integration (OpenSeesPy C++) and neural operator surrogates on Apple Silicon MPS.

1. Benchmark Environment & Methodology

To ensure scientific honesty and reproducibility, timing benchmarks were conducted under strict engineering protocols:

- Hardware Specification:** Apple Silicon Metal Performance Shaders (MPS) unified GPU memory architecture.
- Execution Protocol:** Synchronized process-wide device locks (GLOBAL_DEVICE_LOCK) with Metal command buffer queue synchronization prior to wall-clock recording.
- Benchmark Task:** 5-story building non-linear dynamic response over $T = 20.48$ s (2,048 integration steps, stride 2 evaluation = 1,024 output points).

2. Measured Latency & Throughput Benchmark Results

Model / Solver Implementation	Batch Size	Mean Latency	Throughput	Measured Speedup	Operational Characterization
OpenSeesPy (C++ Numerical Integration)	1	54.68 ms	18.3 sim/s	1.00×	Ground-Truth Reference
EXP6-B (T1-GNO, Native Graph)	1	21.45 ms	46.6 sim/s	2.55× Faster	Recommended Production Surrogate
EXP6-C (Multi-Modal GNO)	1	22.10 ms	45.2 sim/s	2.47× Faster	Multi-Mode Surrogate
EXP6-B (T1-GNO Batch Mode)	32	30.19 ms	1,060.0 sim/s	57.9× (Batch)	Massive Regional Screening
EXP4 (Fixed-Grid FNO-2D)	1	6.60 ms	151.5 sim/s	8.28× Faster	Topology Flawed Baseline

SINGLE-BUILDING REAL-TIME SPEEDUP

2.55×

21.45 ms (GNO) vs. 54.68 ms (OpenSeesPy)

BATCH REGIONAL THROUGHPUT

1,060 sim/s

Batch Size = 32 on Apple Silicon MPS

3. Engineering Context: The Purpose of Surrogate Acceleration

Appropriate Engineering Deployment:

The goal of SeismoFNO is **not to eliminate numerical simulation**, but to enable workflows previously impossible due to computational cost:

- City-Scale Post-Earthquake Screening:** Rapidly screening 50,000 regional buildings in under 60 seconds to identify high-risk structures for emergency response.
- Monte Carlo Seismic Risk Integration:** Convolving thousands of stochastic ground motions over uncertain structural parameters.
- Interactive Structural Design Twin:** Giving engineers instantaneous sub-millisecond slider feedback during preliminary architectural design.

Reproducibility & Independent Forensic Audit

Complete unit test verification, immutable frozen checkpoints, strict partition boundaries, and offline readiness.

1. Automated Test Suite Execution Results

AUTOMATED TESTS PASSING

294 / 294

Full Pytest Verification Suite

TESTS SKIPPED

2

Optional CUDA Hardware Flags

TESTS FAILED

0

100% Pass Rate Across Codebase

2. Independent Forensic Audit Verdict (5/5 Checks Passed)

Prior to documentation release, the repository underwent an automated forensic audit (scripts/run_exp6_forensic_audit.py):

Audit Check	Target Object	Audit Criterion	Verdict
Check 1: Split Independence	Dataset HDF5 Files	Zero sample overlap between ID, OOD-A, OOD-B, OOD-C	PASS (0 Leaked Samples)
Check 2: Checkpoint Immutability	EXP4, EXP5, EXP6 Model Weights	SHA-256 hash match against historical experiment records	PASS (Hashes Frozen)
Check 3: Metric Traceability	Reported Errors & Speedup	100% mathematical match against raw experiment logs	PASS (Fully Traceable)
Check 4: Data Scaler Integrity	Normalizer Transformations	Scalers fit strictly on training split; zero test leakage	PASS (Train-Only Fit)
Check 5: Concurrency Safety	FastAPI Daemon & PyTorch MPS	Process-wide device lock prevents Metal buffer corruption	PASS (10/10 Stress Pass)

3. Complete Provenance Pipeline



Absolute Scientific Freeze Guarantee:

No model was retrained, no dataset was regenerated, and no historical metric was modified during finalization. The system core is fully offline-capable with zero external runtime API dependencies.

Conclusion & Future Research Roadmap

Synthesizing the experimental trajectory and identifying key open directions for computational structural mechanics and AI.

1. The Research Progression Trajectory

STAGE 1 (EXP4): Fixed-Grid FNO Representation Barrier → 3-Story Rel L_2 = 99.60% (Zero-Padding Gibbs Ringing)

↓

STAGE 2 (EXP5): Topology-Native Graph Neural Operator (GNO) → 3-Story Rel L_2 drops to 22.09% (77.51 pp improvement)

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STAGE 3 (EXP5 OOD): Structural Modal-Period Extrapolation Breakdown → Flexible Frame 5S_T120 Peak Error = 35.21% (Rel L_2 = 115.70%)

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STAGE 4 (EXP6): Physics-Informed Modal Invariant FiLM Conditioning → OOD-B Peak Error drops to 13.06% (62.9% relative reduction)

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STAGE 5 (EXP6-D): Falsification Ablation with Shuffled Modal Invariants → Regresses to 24.33% (Validating true physical exploitation)

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STAGE 6 (LIMITATION): Open Scientific Boundary Documented → Static 1D Fourier kernels suffer long-horizon phase drift ($r \approx 0.05\text{--}0.09$)

2. Future Research Directions

1. Time-Frequency Adaptive Wavelet Operators:
Replace global static 1D Fourier kernels with continuous wavelet neural operators (WNO) or multi-scale state-space models (Mamba/S4) capable of dynamically adjusting temporal basis functions to resolve long-horizon phase drift.

2. Phase-Aware Spectral Loss Objectives:
Incorporate dynamic time warping (DTW) and spectral envelope penalties into loss functions to decouple amplitude accuracy from minute phase mismatch during optimization.

3. Conformal Prediction Uncertainty Bounds:
Implement split conformal prediction to provide mathematically guaranteed finite-sample confidence bounds on peak inter-story drift for post-yield safety margins.

4. Irregular 3D & Bi-Directional Topologies:
Expand from 2D planar shear frames to 3D asymmetric building frames with torsional coupling, vertical setbacks, and bi-directional ground motion components ($a_{gx}(t), a_{gy}(t)$).

3. Final Synthesis

“SeismoFNO is therefore best understood as a research prototype demonstrating how structural topology and modal information can be incorporated into neural operators for seismic-response surrogate modeling, while explicitly identifying the remaining generalization limits.”